Notes 3: The GNOME DE and The Bash Shell

Theory Questions

What is a graphical user interface (GUI)?

A GUI is a screen interface with visual elements like icons, windows, buttons, and menus that you can click and interact with using a mouse or touchscreen.

What is a desktop environment?

It's the complete visual package for an operating system. It gives you things like the top bar, application menus, windows, and the background wallpaper (for example, GNOME).

What is the command line interface (CLI)?

A text-based interface where you type out commands line by line and hit Enter to make the computer do things, instead of clicking around with a mouse.

How do I access the command line interface (CLI)?

You can open a terminal window right from your desktop, or switch to a full-screen virtual console using keyboard shortcuts like Ctrl + Alt + F3.

What is a virtual console?

A full-screen, text-only login screen built directly into Linux that runs outside of the graphical desktop. You can jump between a few different ones using shortcuts like Ctrl + Alt + F1 through F6.

What is a terminal emulator?

An app you run inside your desktop (like GNOME Terminal) that gives you a window to type commands into while you're still using your graphical desktop.

What is bash?

Bash is the standard command-line shell (interpreter) used in most Linux distros. It's the program that actually reads what you type and runs it on the system.

What is the shell prompt?

The little chunk of text at the start of your command line (like your username and directory ending in $) that lets you know the computer is ready for your next command.

Command Definitions and Usage

clear

- Definition: Empties out the current terminal screen so you have a clean workspace.

- Usage: clear
- Example: clear

echo

- Definition: Prints out whatever text or string you type after it.
- Usage: echo [text]
- Example: echo "Hello CIS 106"

date

- Definition: Shows the current date and time set on your system.
- Usage: date
- Example: date

free

- Definition: Shows how much RAM and swap memory your system is currently using and how much is left over.
- Usage: free -h
- Example: free -h

uname

- Definition: Displays basic info about your Linux system and kernel version.
- Usage: uname -a
- Example: uname -a

history

- Definition: Shows a list of the previous commands you typed in your current terminal session.
- Usage: history
- Example: history | tail -n 10

man

- Definition: Opens up the official instruction manual for any command.
- Usage: man [command]
- Example: man ls

tldr

- Definition: Gives a quick, easy summary of how a command works with short examples instead of a long manual.
- Usage: tldr [command]
- Example: tldr tar

cheat

- Definition: Displays quick cheat sheets and practical options for Linux commands.

- Usage: cheat [command]
- Example: cheat grep

hostname

- Definition: Displays the name assigned to your computer on the network.
- Usage: hostname
- Example: hostname

df

- Definition: Shows how much disk space is used and available on your hard drive partitions.
- Usage: df -h
- Example: df -h

du

- Definition: Checks how much storage space a specific file or folder is taking up.
- Usage: du -sh [directory]
- Example: du -sh ~

figlet

- Definition: Turns normal text into large, banner-style text made of ASCII characters.
- Usage: figlet [text]
- Example: figlet "Linux"